Dynamic Evidential Routing: An Auditable Neurosymbolic Interface for Clinical Transformers with Incomplete Inputs

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Abstract

Dynamic evidential routing is an inference-time interface for uncertainty-conditioned symbolic intervention. Prototype rules revise a heuristic confidence signal used by a transformer readout. We evaluate 20 hospital-held-out WiDS fits across two imputers, APACHE inclusion/exclusion, and five training seeds, plus 15 clinical-only, genomic-only, and fused TCGA fits using checksum-pinned public mutation files. Every fit uses matched 50-pass dropout inference with three independent repeats and independently replayable intervention exports. The acquired TCGA test cases include 39 recorded BRAF-positive rule events, while absent variants remain unknown without callable assay evidence. Hierarchical comparisons do not meet the prespecified symbolic Brier benefit of 10^{-4}; APACHE removal substantially reduces discrimination. Frozen masking and ensemble diagnostics test implementation behavior on development cohorts. External compatibility, clinical rule validity, and reviewer benefit remain unestablished.

1 Introduction

Clinical AI systems, including emerging generative and agentic systems, need an inspectable way to represent weak evidence and expert intervention. A sparse ICU record may contain a few abnormal measurements, many unmeasured variables, and an apparently precise risk score. Missingness, irregular measurement, and cohort shift are common in hospital data, yet standard neural predictors usually collapse the available evidence into one probability [Li et al., 2019b, Ovadia et al., 2019, Ghassemi et al., 2020]. Any downstream assistant that consumes that score inherits its hidden uncertainty, making an overconfident error difficult to defer, audit, or correct.

The challenge is that current model components expose different pieces of the problem but do not connect them cleanly. Transformers are flexible function approximators for clinical prediction [Rasmy et al., 2021, Azizi et al., 2021, Li et al., 2019a], but attention weights are not themselves evidential confidences. Uncertainty estimators can summarize prediction instability, but they usually do not provide a symbolic intervention point. Rule-based explanations can be readable, but post-hoc rules often describe a prediction without changing the computation. What is missing is an inference-time interface in which neural uncertainty, symbolic evidence, and attention control share an auditable representation.

We propose dynamic evidential routing as that interface. A neural predictive distribution is first mapped into a two-dimensional truth value (f, c), where frequency f describes the balance of evidence and confidence c describes evidential support [Wang, 1995, 2006, Wang and Awan, 2011]. Explicit clinical rules then revise feature-level truths in the same NAL truth-value space, and the revised confidence values control a normalized attention gate. The signal is a heuristic interface

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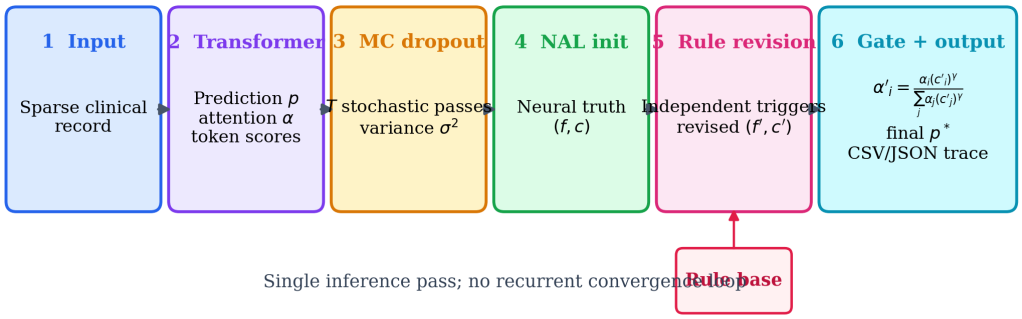


Figure 1: Inference-time dynamic evidential routing. A trained transformer produces a prediction and feature attention; MC dropout estimates uncertainty; the uncertainty signal initializes NAL truth values; triggered clinical rules revise those values; revised confidence gates attention; and the system exports a case-level trace showing where symbolic evidence changed the computation.

variable; its relationship with errors and evidential support must be measured before it can serve as a trust signal.

We evaluate the interface on two complementary clinical benchmarks. Acquired TCGA-THCA mutation files supply recorded held-out positives, but absence of variants is still unknown without verified callable assays. WiDS exercises scale on held-out hospitals with feature-specific missingness; selected-feature missingness is 10.36% after invalid APACHE probabilities are masked. The contribution is an auditable inference-time interface: neural uncertainty and explicit rules share a truth-value representation, revised confidence can change attention, and the arithmetic is exported. The experiments do not establish a clinical performance improvement, and they do not isolate symbolic revision as the source of any calibration difference.

Figure 1 summarizes the inference-time data flow from sparse clinical inputs through uncertainty estimation, NAL-style truth initialization, rule-triggered revision, confidence-gated attention, and trace export.

2 Related Work

Neurosymbolic integration. Neurosymbolic AI combines neural pattern recognition with symbolic reasoning, knowledge injection, or interpretable rule systems [d'Avila Garcez and Lamb, 2023, d'Avila Garcez et al., 2019, Bhuyan et al., 2024, Colelough and Regli, 2025]. Prior surveys distinguish loose coupling, tight coupling, and unified architectures [Hilario, 1997, Nawaz et al., 2025]. Dynamic evidential routing is a tight inference-time interface: the neural model supplies probabilities, uncertainty estimates, and attention weights, while the symbolic layer revises evidential confidence before attention is updated. Its contribution is the interface contract between these components. Neural uncertainty is converted into an explicit truth-value representation, symbolic rules revise that representation, and the revised confidence is fed back into attention with a trace of the intervention.

Clinical transformers and uncertainty. Clinical transformers have been used for structured EHRs, clinical text, and medical prediction tasks [Rasmy et al., 2021, Li et al., 2019a, Dai et al., 2025]. Attention can help interpretability, but medical attention mechanisms rarely represent uncertainty as a first-class control signal [Goncalves et al., 2022]. In parallel, uncertainty quantification methods such as Bayesian neural networks, MC dropout, and ensembles provide useful but imperfect epistemic uncertainty estimates [Blundell et al., 2015, Gal and Ghahramani, 2016, Lakshminarayanan et al., 2017, Lopez et al., 2025]. MC dropout is attractive because it is cheap, but it can be miscalibrated under dataset shift [Ovadia et al., 2019]. Our use of uncertainty is therefore not a standalone calibration method. The uncertainty estimate initializes evidential confidence, which becomes available for symbolic revision and case-level audit.

NARS and biomedical reasoning. NARS and Non-Axiomatic Logic model reasoning under insufficient knowledge and resources [Wang, 1995, 2006, 2022]. NAL truth values have been studied for uncertain medical diagnosis [Wang and Awan, 2011]. Biomedical knowledge graphs and symbolic reasoning can improve clinical interpretability [Wu et al., 2023, Abu-Salih et al., 2023, Rajabi and Kafaie, 2022, Jiang et al., 2024, Chudasama et al., 2025], but many such systems enrich representations rather than exposing an evidential confidence variable that directly controls neural inference. This paper uses NAL truth-value functions and rule-triggered revision only. It does not implement full NARS concept memory, task scheduling, resource allocation, or evidential-base management.

The closest families differ at the intervention boundary. Ordinary attention gating has no separately inspectable symbolic judgment; uncertainty calibration changes confidence without a rule-level intervention trace; post-hoc rules describe a completed decision; and training-time knowledge injection generally requires retraining to change the symbolic source. Dynamic evidential routing instead gives uncertainty and rules a shared truth-value interface, lets rules change inference in one pass, and preserves the source and arithmetic of that change. It does not claim a new general logic or a complete NARS architecture.

3 Method

3.1 Interface Boundary

The implementation should be read as a NAL truth-value interface inspired by NARS, not as a full NARS system. It uses the (f, c) truth-value representation, deduction-style grounding of triggered rules, and revision of neural and symbolic judgments. It does not maintain concept memory, task queues, priority dynamics, resource-bounded control, or explicit evidential-base objects. The claims in this paper are therefore interface-level claims about truth-value-guided fusion and attention control.

3.2 Truth Values

For a statement S with positive evidence w^+ and negative evidence w^-, NAL represents truth as (f, c):

$$f = \frac{w^+}{w^+ + w^-}, \tag{1}$$

$$c = \frac{w^+ + w^-}{w^+ + w^- + k}. \tag{2}$$

We use the standard default $k = 1$, so $c = W/(W + 1)$ for total evidence $W = w^+ + w^-$. Strong deduction combines two premises by

$$f = f_1 f_2, \qquad c = f_1 c_1 f_2 c_2. \tag{3}$$

Revision merges two judgments with weights $\omega_i = c_i/(1 - c_i)$:

$$f = \frac{\omega_1 f_1 + \omega_2 f_2}{\omega_1 + \omega_2}, \qquad c = \frac{\omega_1 + \omega_2}{\omega_1 + \omega_2 + 1}. \tag{4}$$

Standard NAL revision assumes disjoint evidential bases [Wang, 2025]. The two software paths share clinical observations, so disjoint evidential bases are not established. Neural feature frequency is attention whereas symbolic frequency describes a clinical proposition; their semantic compatibility is unvalidated. Revised frequency is exported but does not affect the confidence-only gate.

3.3 Neural-to-Symbolic Initialization

For a neural probability p and MC-dropout predictive variance σ^2, the initial truth value is

$$f_{\text{neural}} = p, \qquad c_{\text{neural}} = \frac{p(1-p) + \varepsilon}{p(1-p) + \sigma^2 + \varepsilon}. \tag{5}$$

This is an application-specific initializer, not a canonical NARS measurement of evidential support. It parallels a variance-controlled evidence mass, then lets symbolic rules revise the confidence in NAL space. If the variance estimator underestimates uncertainty, c can be inflated. This limitation is inherited from the neural uncertainty method.

3.4 Symbolic Rule Revision

Each rule is a clinical proposition with a deterministic condition and a pre-specified symbolic truth value. A rule "fires" exactly when its condition evaluates true for the case. Firing does not depend on $(f_{\mathrm{sym}}, c_{\mathrm{sym}})$: f_{sym} encodes the prototype balance of supporting evidence and c_{sym} controls its evidential weight during revision. Triggered rules independently revise their matching feature truths by Eq. (4); any subset of rules may fire, and no conjunction of all four rules is required. The thresholds are interface probes, not learned decision thresholds or validated clinical cut points.

The symbolic truth values are prototype prior strengths, not an authoritative clinical knowledge base or literal outcome probabilities. A fired rule produces an inspectable event containing its identifier, condition, source feature, neural truth, symbolic truth, revised truth, and attention effect. A domain expert can therefore locate where a prior entered the model and can edit or disable that prior without retraining the transformer.

3.5 Attention Gating

Let α_i be the original attention weight for feature i and c_i the revised confidence associated with that feature. The gated attention is

$$\alpha_i^{\text{new}} = \frac{\alpha_i c_i^{\gamma}}{\sum_j \alpha_j c_j^{\gamma}}, \tag{6}$$

where γ controls the strength of confidence gating. The update preserves normalization and sets a feature's contribution to zero when its confidence is zero. In this paper, $\gamma = 2$ is the default; a sensitivity run over $\gamma \in \{0.25, 0.5, 1, 2, 4\}$ is reported for TCGA.

3.6 Inference-Time Data Flow

The inference path has six stages. First, the trained transformer produces a point prediction, feature attention, and token scores. Second, repeated MC-dropout passes estimate predictive and attention variance. Third, the variance-to-confidence initializer in Eq. (5) creates feature-level neural truths. Fourth, the symbolic extractor emits every proposition whose condition is true. Fifth, neural and symbolic truths are revised in the shared (f, c) space. Sixth, revised confidence gates attention by Eq. (6); current implementation recomputes each cached pass's final logit and averages sigmoid probabilities identically across gate variants, then exports all events. The default intervention changes the token readout, not encoder representations; a separate encoder intervention experiment recomputes those representations. There is no recurrent feedback or convergence loop.

This staged design keeps the symbolic mechanism outside model training. With N cases, R rule predicates, and F features, predicate evaluation and gating cost $O(NR + NF)$ in the present vectorized implementation. The prototype assigns at most one rule to a logical feature and rejects collisions, so rule ordering cannot silently change a truth value. Larger bases need provenance-aware conflict policies, coverage analysis, and explicit aggregation of compatible or contradictory evidence before same-feature multi-rule revision can be supported.

3.7 Rule-Trace Semantics

For every triggered rule, the implementation records the feature name, rule condition, neural truth value, symbolic truth value, revised truth value, and resulting attention gain. These traces are not merely explanations after the fact. They correspond to actual arithmetic used by the prediction pipeline. A rule trace can therefore support three kinds of audit: whether the right rule fired, whether its symbolic truth value is clinically defensible, and whether the revised confidence changed attention in the expected direction.

This is the main neurosymbolic distinction from a post-hoc explanation. In a post-hoc explanation, symbolic language may describe a neural decision without changing the computation. In dynamic evidential routing, symbolic truth values become part of the computation itself. The trace can still be wrong if the rule base is wrong or the proposition extractor is misaligned, but the failure is localizable.

3.8 Worked Trace

For a neural estimate $p = 0.8$, $\sigma^2 = 0.05$, and a fired prototype rule $(f_{\text{sym}}, c_{\text{sym}}) = (0.9, 0.7)$, Eq. (5) gives $(f_{\text{neural}}, c_{\text{neural}}) \approx (0.8000, 0.7619)$. Revision gives $(f_{\text{rev}}, c_{\text{rev}}) \approx (0.8422, 0.8469)$. In natural language, the rule contributes relatively strong positive evidence with moderate confidence; it does not mean a clinician assigned a 0.9 patient-level risk.

4 Experimental Setup

The current development evaluation fixes source hashes, partitions, rules, and training settings before each fit. WiDS contains 91,713 labeled stays with 65,737 training, 11,751 validation, and 14,225 held-out cases; patient, hospital, and ICU identifiers are disjoint across partitions. Five training seeds are evaluated under each combination of KNN or median imputation and APACHE included or excluded, yielding 20 fits. Removing APACHE leaves 15 encoded features instead of 16; score-only comparators remain available as controls.

TCGA contains 457 labeled cases with 319 training, 69 validation, and 69 held-out cases. An immutable GDC manifest pins 498 mutation files by publisher UUID, checksum, and case coverage; 16 expected clinical cases lack file-associated coverage. The fused input has 55 clinical and 100 genomic/unknown-indicator dimensions. Clinical-only, genomic-only, and fused inputs each receive five training seeds with identical partitions. Unknown mutation calls are not converted to negatives; no complete callable panel is verified. This is retrospective post-pathology association, and source-level measurement chronology is unresolved.

All fits use $d_{\text{model}} = 64$, four heads, two encoder layers, at most 60 epochs, and validation early stopping with patience 12. Each fit has 50 matched MC-dropout passes and three independent inference repeats. Calibration and classical-model selection use validation data only. These already inspected cohorts constitute development evaluation, not prospective confirmation or independent-cohort validation.

4.1 Baselines and Controls

The baseline transformer tests whether the underlying neural architecture is competitive before confidence gating is introduced. A uniform normalized readout gate cancels mathematically. Under matched current implementation aggregation it must reproduce ungated predictions within 10^{-7}. The legacy flat-control differences arise from aggregation, not from adding a gate. The MC-confidence-only ablation uses the neural variance-derived confidence but does not apply symbolic revision. The NARS-gated model adds the symbolic rule path on top of the MC-confidence initializer. This ordering is important: a difference from the baseline alone is not enough to show that symbolic reasoning helped, because confidence gating or regularization may explain the change.

The random forest baseline addresses the common tabular-learning concern that tree ensembles can be strong competitors on structured clinical features. It is trained on the same split and post-preprocessing design matrix, so differences cannot be attributed to a different data representation.

4.2 Symbolic Isolation, Uncertainty, and Shift

Tables 5 and 6 do not isolate symbolic revision. The evaluation contract now rescores NAL revision after four changes to the symbolic input: removing every rule, removing one rule at a time, permuting rule assignments across patients, replacing symbolic truth values with random draws, and substituting fixed symbolic priors. Rule removal must reproduce MC-confidence-only gating. Sensitivity scales symbolic confidence and recomputes revision; it does not retune the transformer on test labels. A claim that the symbolic component improves prediction would require the stated rules to separate from MC-confidence-only gating and from these controls. That separation is not a result of the saved runs.

Each condition now has a five-model ensemble, MC-dropout variance, predictive entropy, confidence-error diagnostics, and tie-aware selective-risk curves. Common ensemble prediction errors provide the same reference for comparing uncertainty estimators. Hospital, training-seed, and dropout-repeat resampling distinguish these sources of variability; the fixed split does not estimate variability across cohort selections.

Table 1: Prototype symbolic rules. Conditions are evaluated independently; any subset may fire. Truth values are evidential weights for the interface, not validated clinical probabilities.

Data	Rule	Trigger condition	(f, c)	Clinical reading
TCGA	BRAF mutation	BRAF mutation indicator = 1	$(0.85, 0.75)$	The rule increases support for the BRAF feature when the mutation is recorded; it is not a validated risk probability.
TCGA	Age >= 55	age at diagnosis >= 55 years	$(0.70, 0.60)$	The rule raises evidential support for the age feature without asserting that age alone determines metastasis.
TCGA	Pathologic T3/T4	pathologic T category is T3, T3a, T3b, T4, T4a, or T4b	$(0.90, 0.85)$	The rule increases support for the observed staging feature; it does not establish nodal spread by itself.
TCGA	Extrathyroid extension	extrathyroid extension is Minimal (T3), Moderate/Advanced (T4a), or Very Advanced (T4b)	$(0.85, 0.80)$	The rule raises support only for an explicitly recorded extension category; missing, placeholder, and unknown values do not fire.
WiDS	Lactate	maximum day-1 lactate >= 4.0 mmol/L	$(0.85, 0.80)$	The rule increases support for the lactate feature without diagnosing shock or determining mortality.
WiDS	Hypotension	minimum day-1 systolic blood pressure <= 90 mmHg	$(0.75, 0.70)$	The rule increases support for the blood-pressure feature; it is not a stand-alone mortality decision.
WiDS	Age	age >= 75 years	$(0.65, 0.60)$	The rule increases support for the age feature without treating age as sufficient for mortality.
WiDS	Creatinine	maximum day-1 creatinine >= 2.0 mg/dL	$(0.70, 0.65)$	The rule increases support for the creatinine feature without diagnosing kidney injury or determining mortality.

Each rule now carries a source note, a provenance identifier, an expert-review status, and a contradiction group. The current predicates are prototype probes and are marked not reviewed. Two rules that share a contradiction group and a target feature are rejected rather than overwritten. No independent cohort is evaluated here. The distribution-shift study masks originally observed raw values at 0, 10, 30, 50, and 70 percent before frozen preprocessing, rule extraction, and matched inference. Random, feature-dependent, and explicitly labeled outcome-dependent simulations share masks across methods. That check is a compatibility test under shift, not external clinical validation.

4.3 Rule Bases

The TCGA rule base contains four thyroid-cancer rules chosen to exercise distinct input channels: BRAF mutation for genomic evidence, age at least 55 years for demographic evidence, pathologic T3/T4 stage for staging evidence, and extrathyroid extension for pathology-derived evidence. The WiDS rule base contains four ICU rules chosen to exercise common tabular critical-care signals: lactate for metabolic stress, hypotension for hemodynamic instability, age for demographic risk, and creatinine for renal dysfunction. All rules are applied only at inference time.

The small rule bases are prototype probes, not complete clinical ontologies. Clinical use would require expert-curated propositions, population-specific threshold rationales, provenance, and sensitivity analysis. Existing guideline or score components could be encoded and run in shadow mode to measure coverage, disagreement with extracted data, and downstream model effects. Such experiments could reveal where a rule interacts with the model; they would not validate the clinical correctness of the rule itself. Symbolic revision would be empirically isolated only if it separated from baseline, flat-confidence, and MC-confidence controls in paired comparisons. The current results do not meet that standard.

4.4 Statistical Reporting

The metric tables report point estimates followed by 95% bootstrap confidence intervals over held-out examples for AUC, Brier score, and ECE. Accuracy is reported as a point estimate. These intervals are confidence intervals, not standard deviations. This matters because the transformer variants are extremely close numerically. A table entry that is best at the fourth or fifth decimal place should not be interpreted as a confirmed improvement unless a corresponding interval or paired comparison excludes zero.

The paper therefore separates three evidential claims. First, whether the architecture compiles and runs on the benchmark. Second, whether the symbolic pathway fires and changes feature-level truth values. Third, whether the symbolic pathway changes predictive metrics beyond simpler controls. The present results support the first two claims. They do not isolate the third claim. This separation is central to the paper's framing.

Table 2: Current development evaluation: five training seeds per condition on fixed held-out cases. Brier values are means with seed standard deviations, not case-bootstrap confidence intervals. The APACHE score remains a score-only comparator when removed from predictor inputs.

Condition	Variant	AUROC	Brier (seed SD)
WiDS KNN	Ungated	0.875721	0.057830 (0.000109)
WiDS KNN	MC-only	0.875711	0.057828 (0.000112)
WiDS KNN	NARS	0.875713	0.057828 (0.000112)
WiDS KNN	Boosting	0.873278	0.057864 (0.000156)
WiDS KNN	Recalibrated	0.875721	0.057782 (0.000145)
TCGA fused	Ungated	0.692941	0.229410 (0.009980)
TCGA fused	MC-only	0.692101	0.229534 (0.010085)
TCGA fused	NARS	0.692101	0.229548 (0.010068)
TCGA fused	Boosting	0.661345	0.243924 (0.000000)
TCGA fused	Recalibrated	0.692941	0.226140 (0.008783)

Table 3: Matched symbolic contrasts, NARS-minus-MC Brier, with hierarchical 95% intervals over cases/hospitals, training seeds, and three dropout repeats. Negative is better. These development comparisons do not meet the prespecified 10^{-4} minimum benefit.

Condition	Difference	Lower	Upper
WiDS KNN	-2.43e-07	-6.91e-07	1.64e-07
WiDS median	-4.04e-07	-1.19e-06	5.79e-07
WiDS KNN, no APACHE	-1.68e-07	-4.31e-07	3.11e-07
WiDS median, no APACHE	-2.35e-07	-7.85e-07	6.53e-07
TCGA fused	1.26e-05	-5.69e-05	1.06e-04
TCGA clinical	2.63e-05	-5.74e-05	1.33e-04
TCGA genomic	-1.04e-05	-4.35e-05	8.06e-06

4.5 Reproducibility Interface

The public repository exposes a single entrypoint for running both dataset branches and producing dataset-scoped metrics, traces, plots, and an aggregate run summary. The paper text reports the split sizes, input dimensions, architecture, MC-dropout count, and baselines needed to interpret the results. The code is still necessary for exact reproduction because preprocessing, imputation, categorical encoding, and rule extraction are implementation-sensitive.

In current implementation, imputation, standardization, categorical encoding, sparse-column filtering, and mutation-panel selection fit training data only. Legacy sparse filtering and panel selection occurred before splitting. Unknown gene calls now have explicit missing indicators; confirmed negative calls require sourced case/gene assay coverage. TCGA selects one physical record per source table and exports conflicts, but cross-table chronology is unverified; its landmark is retrospective post-pathology association, not preoperative prediction. The rule extractor is also run only at inference time, so symbolic rules do not alter the supervised training objective.

5 Current Development Results

The seven conditions in Table 3 do not establish the planned symbolic benefit. WiDS KNN NARS-minus-MC Brier is -2.43×10^{-7} with hierarchical 95% interval $[-6.91\times10^{-7}, 1.64\times10^{-7}]$. TCGA fused routing has a positive mean difference of 1.26×10^{-5}, with an interval spanning zero. Direct confidence boosts, frequency sensitivity, individual/all-rule removal, 100 predicate permutations, shuffled confidence assignments, and matched irrelevant predicates are retained as symbolic controls. Revised frequency does not determine the confidence gate, and shared input measurements do not establish independent evidence.

Mean WiDS ungated AUROC is 0.875721 with KNN and APACHE, versus 0.841118 without APACHE; median yields 0.874062 versus 0.842176. Paired hospital-and-seed contrasts preserve common test cases and report multiplicity-adjusted intervals. Boosting beats the median transformer in mean Brier, and validation recalibration can exceed the small routing effects. These results expose dependence on an existing mortality score rather than establish added clinical value over the strongest comparator. Authoritative APACHE sentinel semantics and timestamp availability remain unverified.

Table 4: Operational auditability for each training seed on the same held-out cohort. Events include unmapped triggers; finite values and confidence changes refer to mapped events. Rows repeat cases across fits and must not be added. Residuals measure numerical fidelity, not clinician usability or rule validity.

Data	Cases covered	All (mapped)	Finite	c changed	Flips	Max residual
TCGA-THCA seed_0	56/69 (81.2%)	113 (89)	100.0%	100.0%	0	4.4e-16
TCGA-THCA seed_1	56/69 (81.2%)	113 (89)	100.0%	100.0%	0	4.4e-16
TCGA-THCA seed_2	56/69 (81.2%)	113 (89)	100.0%	100.0%	2	4.4e-16
TCGA-THCA seed_3	56/69 (81.2%)	113 (89)	100.0%	100.0%	0	4.4e-16
TCGA-THCA seed_4	56/69 (81.2%)	113 (89)	100.0%	100.0%	0	4.4e-16
WiDS ICU seed_0	8,152/14,225 (57.3%)	11,457 (11,457)	100.0%	100.0%	3	3.3e-16
WiDS ICU seed_1	8,152/14,225 (57.3%)	11,457 (11,457)	100.0%	100.0%	2	4.4e-16
WiDS ICU seed_2	8,152/14,225 (57.3%)	11,457 (11,457)	100.0%	100.0%	3	4.4e-16
WiDS ICU seed_3	8,152/14,225 (57.3%)	11,457 (11,457)	100.0%	100.0%	0	4.4e-16
WiDS ICU seed_4	8,152/14,225 (57.3%)	11,457 (11,457)	100.0%	100.0%	1	4.4e-16

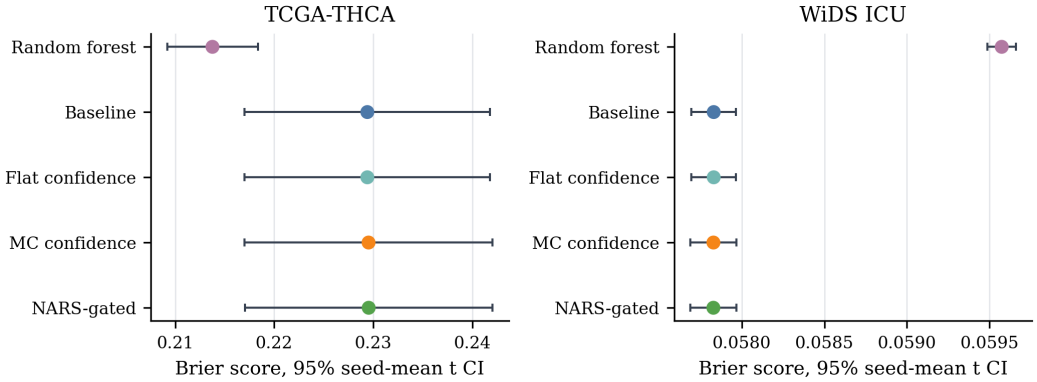


Figure 2: Current five-seed mean Brier scores with 95% Student-t intervals over training seeds on fixed held-out cases. These are seed-mean intervals, not case-bootstrap intervals or external confirmation.

The acquired TCGA held-out panel has recorded positives in 50 of 69 cases and produces 39 BRAF, 29 T-stage, 24 extension, and 21 age events per seed. All 113 events are exported; 89 map to fitted features and 24 are explicitly unmapped. WiDS observed-only rules produce 11,457 mapped events in 8,152 of 14,225 cases per seed. Numerical replay validates predicates, truth revision, gates, probabilities, and rule-off effects; it does not certify the clinical predicates. Seed rows reuse the same cases and must not be combined as independent patient counts.

Controlled missingness outputs preserve frozen preprocessing, rules, model weights, and matched dropout inference. Per-fit paired intervals test integrated degradation and the 0.001 per-mask harm margin; uncertainty diagnostics include error-detection AUROC and selective-risk area with hospital-clustered intervals. The criterion passes in 12 of 15 median seed/scenario combinations and 1 of 15 KNN combinations; no KNN random or feature-dependent scenario passes. This does not establish general routing robustness. Outcome-dependent masks are simulations, not observed missingness mechanisms, and numerical acceptance under a particular simulation does not establish population robustness.

For the fixed five-model WiDS KNN ensemble, mean predictive MC variance detects common threshold errors with AUROC 0.842276 and hospital-bootstrap interval [0.833229, 0.857306]. Neural and revised attention-instability scores give 0.514950 [0.493533, 0.531916] and 0.500647 [0.477047, 0.515520], respectively. Their intervals span chance: informative predictive uncertainty does not establish the attention confidence as a useful trust signal. These conditional intervals exclude model-refitting variability.

The reviewer protocol, balanced format assignments, investigator fault key, power sensitivity, and crossed reviewer/case analysis are implemented as a demonstration package. No participants were recruited and no human benefit is claimed. Its public fault key requires regeneration into a private study

Table 5: Held-out TCGA-THCA performance for lymph node metastasis prediction. Point estimates are followed by 95% bootstrap confidence intervals where shown.

Variant	AUC	Brier	ECE	Acc.
Random forest	0.6613 [0.5101, 0.7908]	0.2200 [0.1895, 0.2498]	0.1597 [0.1317, 0.2824]	0.5942
Baseline transformer	0.7311 [0.5975, 0.8462]	0.2113 [0.1825, 0.2426]	0.1184 [0.1074, 0.2517]	0.6812
Flat-confidence	0.7328 [0.6028, 0.8479]	0.2109 [0.1816, 0.2427]	0.1178 [0.1130, 0.2641]	0.6812
MC-confidence-only	0.7311 [0.6008, 0.8453]	0.2110 [0.1820, 0.2418]	0.1184 [0.1041, 0.2481]	0.6812
NARS-gated	0.7311 [0.6008, 0.8453]	0.2109 [0.1820, 0.2417]	0.1184 [0.1042, 0.2481]	0.6812

Table 6: Held-out WiDS ICU performance for hospital mortality prediction. Point estimates are followed by 95% bootstrap confidence intervals where shown.

Variant	AUC	Brier	ECE	Acc.
Random forest	0.8768 [0.8669, 0.8868]	0.05817 [0.05520, 0.06142]	0.00737 [0.00596, 0.01254]	0.9258
Baseline transformer	0.8803 [0.8700, 0.8902]	0.05653 [0.05371, 0.05966]	0.00750 [0.00493, 0.01211]	0.9285
Flat-confidence	0.8802 [0.8698, 0.8901]	0.05648 [0.05366, 0.05964]	0.00618 [0.00421, 0.01098]	0.9286
MC-confidence-only	0.8802 [0.8698, 0.8901]	0.05647 [0.05365, 0.05962]	0.00581 [0.00409, 0.01062]	0.9286
NARS-gated	0.8802 [0.8698, 0.8901]	0.05647 [0.05365, 0.05962]	0.00583 [0.00411, 0.01064]	0.9286

package before real recruitment. External evaluation similarly requires a documented independent cohort with matched units, windows, outcomes, and patient identities; no such cohort was supplied.

6 Legacy Results and Validity Audit

The two numerical tables in this legacy section describe the archived implementation; current performance figures and generated auditability tables are reported separately above. They use unmatched prediction aggregation and an internal row split. Only two TCGA training cases have recorded mutation inputs; validation and test contain none. No genomic held-out intervention was demonstrated. Close point estimates establish neither equivalence nor noninferiority. The short integration artifacts remain software checks; the full development studies above replace the historical runs as current numerical evidence without turning them into prospective confirmation.

On TCGA-THCA, the transformer variants remain tightly clustered. Flat-confidence has the highest AUC at 0.7328 and the lowest Brier point estimate at the displayed precision; the 69-case test split does not statistically separate variants. TCGA therefore provides a limited clinical interface exercise; it does not establish held-out multimodal feasibility or symbolic-gating superiority.

The legacy TCGA clinical rules were active, but the BRAF rule never fired on test. Counts and coverage below use only recognized extrathyroid-extension categories; placeholder and unknown values cannot fire.

WiDS provides the clearer scale test. All transformer variants have higher AUC point estimates than the random forest in this run, although their AUC confidence intervals overlap. Within the transformer family, the baseline has the highest AUC, while MC-confidence-only has the numerically lowest Brier and ECE point estimates. NARS-gated is nearly identical to MC-confidence-only and ties it and flat-confidence on accuracy at the reported precision.

The archived intervals below describe the earlier calculation. Figure 2 instead uses the explicitly selected current five-seed sources, so its values must not be compared to these legacy intervals as a controlled experiment.

The symbolic pathway operated at scale: ICU rules fired on 8,551 of 13,757 held-out stays, with 13,031 feature-level revisions. The per-rule counts were 1,936 for lactate, 5,338 for hypotension, 3,443 for age, and 2,314 for creatinine. Every fired rule mapped to a feature, produced a finite trace, and changed confidence and attention. The corresponding TCGA rates were also 100% across 79 events in 42 of 69 cases. The legacy zero revision residual reused production arithmetic and was a consistency check. Its finite in-memory values did not establish complete persisted event export. WiDS counts included imputed-value triggers; current implementation suppresses these and independently replays every exported event.

6.1 Case-Level Rule Trace

The legacy prediction export provides a concrete held-out example. For WiDS encounter 8205 (target 0), the ungated probability was 0.5006, flat-confidence gating produced 0.4995, MC-confidence-only gating produced 0.4961, and NARS-gated routing produced 0.4968. Three rules fired: hypotension, lactate, and creatinine. Their attention weights changed from 0.2564 to 0.2526, 0.0231 to 0.0248, and 0.0347 to 0.0372, respectively. The trace therefore exposes a real threshold change and the rule-level arithmetic, but it also shows why attribution requires controls: MC-confidence-only had already crossed the threshold before symbolic revision. Across the complete WiDS export, MC-confidence-only and NARS-gated produced no different threshold decisions. These legacy decision changes describe the entire inference procedure, not an isolated symbolic effect or clinical validation.

6.2 Calibration Interpretation

The legacy WiDS calibration differences are descriptive and confounded by aggregation. Relative to the ungated transformer, NARS-gated routing lowers Brier by 0.0000587 with a paired 95% interval of [0.0000249, 0.0000911], and lowers ECE by 0.00120 with an interval of [0.0000282, 0.00202]. Both intervals exclude zero. Relative to flat-confidence and MC-confidence-only gating, however, the paired Brier and ECE intervals include zero; MC-confidence-only also has the slightly better Brier and ECE point estimates. The baseline and gated predictions use different aggregation, so even a confidence-gating benefit cannot be isolated from this run. The legacy reported deltas are bootstrap means rather than the required observed effect sizes; the current implementation exporter distinguishes them.

That restraint keeps the contribution aligned with what the experiment actually supports. Dynamic evidential routing should not be described as a universal calibration booster. The supported implementation claim is that explicit rules can change a heuristic confidence readout. Robustness, predictive compatibility, and useful auditability require corrected experiments.

7 Discussion and Ablation

The flat-confidence and MC-confidence-only controls are essential for interpreting the result. If NARS-gated attention were compared only with the baseline transformer, its paired Brier and ECE differences could be over-interpreted as a symbolic reasoning effect. Those controls neither isolate symbolic revision nor establish a gating effect while aggregation differs. The NARS rule path is therefore best interpreted as an instrumentation layer with inspectable numerical activity and editable prototype rules, not yet a statistically isolated source of within-family calibration change.

Figure 3 uses the current five-seed TCGA fused and WiDS KNN sources. Gamma sweeps and decision curves are descriptive development diagnostics; no gamma is selected using test outcomes, and overlapping net-benefit curves do not establish clinical utility.

7.1 Failure Modes and Deployment Boundary

The interface can fail when MC dropout underestimates uncertainty, extraction misaligns features with clinical concepts, or rule truth values are disputed. Traces expose extracted conditions and revisions for inspection and rule-off replay, but whether this instrumentation helps human oversight has not been evaluated.

The current system remains a methods prototype rather than a clinical decision device. Deployment would require a rule base curated independently by domain experts, external validation without retuning, shift monitoring, rule governance, human-factors evaluation, and clinician override. Stronger evidence for symbolic revision would also require separation from baseline, flat-confidence, and uncertainty-only controls across external cohorts.

8 Limitations

The inspected development cohorts contain 69 held-out TCGA cases and WiDS hospitals within one benchmark. Both rule bases lack expert review. Ensemble and masking intervals omit cohort-

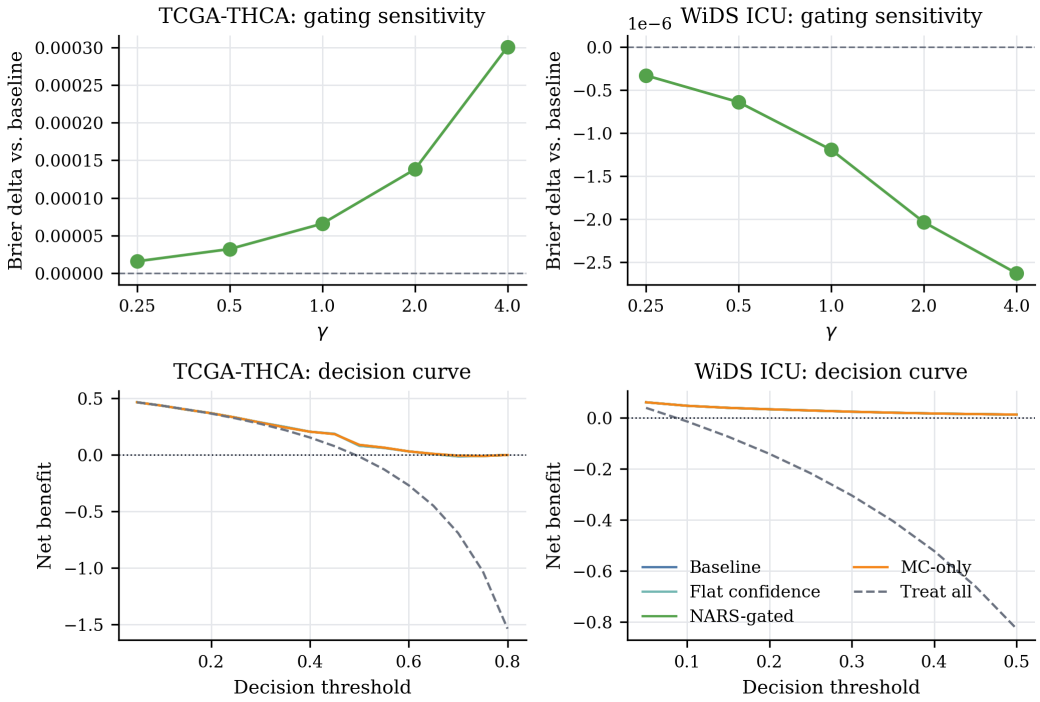


Figure 3: Reported ablations. Top: NARS-gated Brier change relative to the ungated transformer across γ (negative is lower). Bottom: held-out decision curves over informative threshold ranges. These descriptive curves do not establish clinical utility.

selection uncertainty; MC dropout may understate uncertainty under shift. Clinical validation remains unestablished.

An exploratory sensitivity evaluates ten frozen no-APACHE checkpoints on 12,000 PhysioNet/CinC 2012 records[1], with first-day inputs and complete replay. Missing SpO2/surgery inputs, original ≥ 48-hour stay selection, and unverified identity/window equivalence prevent independent clinical claims; intervals condition on one institution. TCGA pathology dates remain absent in 507 selected records. Public GDC biotabs add 515 sample-linked procurement/review offset pairs, but do not verify selected clinical/genomic input availability.

NGTA implements NAL truth-value functions and feature-level revision, without NARS concept priorities, task scheduling, or explicit evidential-base records. Its claims concern an inference interface, not a complete NARS architecture.

9 Conclusion

Dynamic evidential routing is an auditable inference-time interface for uncertainty-conditioned symbolic intervention. Explicit rules can revise confidence and change attention, and the run leaves arithmetic traces. The saved comparisons do not show that symbolic revision improves clinical prediction. External cohorts, expert-curated rules, and human evaluation are needed before clinical use.

Ethics and reproducibility. This study uses existing de-identified benchmarks and collected no new patient data. The WiDS publisher requires registered access and a data-use agreement. An anonymized implementation accompanies the submission. Large language models assisted with drafting, formulation, and code generation; the authors checked the derivations, implementation, references, and results and take responsibility for the work.

[1] Source: `https://physionet.org/content/challenge-2012/1.0.0/`.

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